

Data Classification -- Guided Notes ANSWER KEY

Unit tp-5.1 | CompTIA Network+ N10-009 Obj. FC0-U71 5.1 | N10-008 FC0-U71 5.1

For instructor use / distribute after students complete the notes.

Question 1

Data classification is the process of organizing data into categories based on its _____ and the level of _____ required. Organizations classify data to apply _____ controls -- not all data needs the same protection, and over-protecting low-sensitivity data wastes resources.

Answer: Sensitivity (value); protection; appropriate security.

Question 2

Common civilian data classification levels from lowest to highest sensitivity are: _____ (available to the public), _____ (internal use only, not for external distribution), _____ (limited to authorized staff), and _____ (highest risk if disclosed, limited to need-to-know individuals).

Answer: Public; internal (private); confidential; restricted (top secret / highly restricted).

Question 3

PII (Personally Identifiable Information) is data that can identify a _____ individual. Examples include _____, _____, _____, and _____ (name at least four). Regulations like _____ (US healthcare) and _____ (EU privacy) require specific handling of PII.

Answer: Specific; name, Social Security Number, email address, phone number, address, DOB (any four); HIPAA; GDPR.

Question 4

PHI (Protected Health Information) is a specific subset of PII relating to a person's _____ condition, treatment, or payment for care. It is protected by _____ in the United States. Organizations that handle PHI (hospitals, insurers, labs) are called _____ entities and must follow strict safeguards.

Answer: Health (medical); HIPAA; covered.

Question 5

The principle of _____ requires that individuals and systems only have access to the data they need for their specific job function -- no more. A marketing employee should not have access to _____ records. A help desk technician does not need access to the company's _____ source code repository.

Answer: Least privilege; financial (payroll); proprietary (development).

Question 6

Data _____ marks files with their classification level so access control policies can be enforced automatically. _____ prevents sensitive data from leaving an organization via email, USB, or cloud upload by monitoring and blocking classified content. This technology is called _____ (Data Loss Prevention).

Answer: Labeling (tagging); technology; DLP.

Question 7

Structured data is organized in a defined format that can be easily _____ (e.g., database tables, spreadsheets). Unstructured data has no predefined format -- examples include _____, _____, and _____. Semi-structured data has some organizational properties but does not fit a relational model -- _____ and XML are examples.

Answer: Searched/queried; emails, images, videos (any two); JSON.

Question 8 -- Real World Application

REAL WORLD: A hospital employee accidentally emails a spreadsheet containing 500 patients social security numbers and diagnoses to an external vendor. Identify the regulatory framework violated, classify the data by type (PII/PHI), and describe two technical controls that could have prevented this from happening.

Answer: Regulatory violation: HIPAA (Health Insurance Portability and Accountability Act). The data contains both PII (SSNs, which identify individuals) and PHI (diagnoses combined with identity -- patient health information). HIPAA requires covered entities to protect PHI and report breaches affecting 500 or more individuals to HHS and the media. Technical control 1 -- DLP (Data Loss Prevention): a DLP system scans outbound emails for patterns matching SSN formats and PHI keywords (e.g., ICD codes, diagnosis terms). It would automatically block the email and alert the security team before delivery. Technical control 2 -- Data classification and access controls: implement role-based access so clinical staff data exports require authorization and SSN columns are masked or removed for non-clinical workflow exports. The vendor may only need aggregate statistics, not individual identifiers.